Microsoft Windows Server System
Customer Solution Case Study

University Uses Existing Windows Investments to Tighten Network Security and Management

Overview

Country or Region: Brazil
Industry: Education

Customer Profile

Universidade de Vila Velha (UVV) is a well-known university in Brazil, offering degree programs to more than 14,000 students in many disciplines, including medicine, engineering, the sciences, and law.

Business Situation

The IT staff at UVV operated two physically separate computer networks—one for students and one for administration. But the networks were costly to maintain and difficult to secure and manage.

Solution

Using a Server and Domain Isolation solution based on Internet Protocol security (IPsec), UVV was able to consolidate two separate networks and increase network security without any additional hardware or software investment.

Benefits

- Improved security
- Lower operating cost that facilitates growth
- Improved productivity

"Server and Domain Isolation is an amazing solution. We already had all the tools .... Once we had time to study and to plan the IPsec solution, we did it quickly ... and at no additional cost."

Rodrigo Immaginario, Chief Information Officer, Universidade de Vila Velha

The Universidade de Vila Velha (UVV) in Brazil is host to 14,000 students. Like most universities, UVV provides computer resources for its students, faculty, and administrative staff. To ensure better security, UVV kept its faculty and staff network separate from the student network. But managing separate networks was difficult and became increasingly expensive as the university continued to grow. In 2005, UVV implemented a highly secure network using a Server and Domain Isolation solution based on Internet Protocol security (IPsec). The solution allowed the school's two networks to coexist on a single physical network. As a result, UVV was able to increase security networkwide, safeguard intellectual property, and simplify network management, thereby increasing IT staff productivity—all at no additional hardware or software expense to the university.

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Situation

Founded in 1976, Universidade de Vila Velha (UVV) is one of Brazil's most renowned centers of higher learning. The 600-member faculty teaches about 14,000 students in various disciplines, including medicine, engineering, and law. The university also employs about 300 administrative personnel. Vila Velha, a well-known tourist destination on the eastern coast of Brazil, is home to the main campus of UVV. The university also operates campuses in three other locations.

In most university environments, the administrative staff, faculty, and students need access to computer resources. Managing security in this type of an environment is always a challenge for IT groups. At UVV, the IT group chose to secure its environment by designing the faculty and staff network to be physically separate from the student network. This required separate cabling and network switches as well as separate servers and peripheral devices.

The server computers at UVV run the Microsoft® Windows Server™ 2003 operating system with the Active Directory® service, part of Microsoft Windows Server System™ integrated server software. Students and lab resources belong to one Active Directory domain; faculty and staff belong to another. To safeguard the faculty and staff network resources, the university did not allow any direct communication between the faculty and staff network and the student network, so there was no sharing of print, file, or application resources.

UVV also had intellectual property to safeguard. The IT group at UVV had developed a comprehensive management application that the university uses to manage its own administrative operations. That application has been so successful that UVV now sells the solution to other universities. The systems that store the source code reside in the faculty and staff network, but the IT group wanted to strengthen the security boundary of these systems to safeguard the university's intellectual property.

As the university continued to grow rapidly, managing two physically separate networks became increasingly challenging for the 22-member IT staff. "The existing architecture of separate networks for students and administration was too expensive to maintain and too difficult to manage; it didn't allow us to grow without incurring high costs," says Rodrigo Immaginario, who is Chief Information Officer at UVV and also a Microsoft Most Valuable Professional (MVP). "Our challenge was to introduce one technology that would allow these networks to coexist on a single physical network while ensuring the same level of security and keeping the cost low."

For Immaginario, it was important that the IT group find a solution that could be implemented in a very short period of time—just over two weeks in late December. He wanted a solution that would not disrupt administrative operations and would be completely transparent to users.

Solution

In early December 2005, the IT group at Universidade de Vila Velha began working with Microsoft to redesign the UVV network infrastructure using a Server and Domain Isolation solution based on Internet Protocol security (IPsec) and the Active Directory service. Using this solution, administrators can logically isolate managed domains and create secure virtual networks for specific servers, such as the UVV systems that store the source code for the university's management application. Administrators manage access between isolated domains using Active Directory Group Policy settings; they use IPsec to enforce those policies. IPsec, which is fully integrated with Active Directory,

operates transparently at the network level to authenticate the origin and integrity of data and, if needed, provide confidentiality for data. And IPsec requires no modification of existing applications or services.

To implement a Server and Domain Isolation solution using IPsec, the IT group had to first identify which systems were trusted and which were not. At UVV, only systems and users that are part of the university's Active Directory structure are considered trusted; all systems and users that are outside of the university's Active Directory structure, such as the student network, are not trusted. Only trusted systems are allowed to connect to the network domain servers. All inbound traffic requires IPsec authentication, and only trusted systems can initiate connections.

UVV uses Domain Isolation to configure which client computers are authorized to connect to specific server systems. For example, students are only authorized to connect to resources on the student network. In turn, domain server systems are configured to accept connections only from client systems that can authenticate traffic using IPsec.

To further safeguard the UVV application source code, the IT group uses a more restrictive Server Isolation to shield the critical servers and data stores on which the UVV source code resides. "Only developers and myself are able to connect to the servers that run and store the UVV application source code," says Immaginario. "As an added layer of protection, we also use IPsec to encrypt all traffic to and from the source code servers."

Because the solution required no changes to the UVV Active Directory structure, the staff was able to implement the solution in record time. It took just 2 days to secure about 1,000 desktops and 30 servers distributed across four campuses. "We spent about 10 days in December reviewing all the requirements, scenarios, solutions, systems, and applications, and then mapped out this information," says Immaginario. "Then, in January 2006, we were able to deploy the solution in just 2 days with little interruption of service. The solution was completely transparent to our users."

Benefits

By using IPsec and Group Policy to implement a Server and Domain Isolation solution, Universidade de Vila Velha achieved its goal of securing its two existing networks at a low cost—and it did so in record time. Network security is now better than it was before; productivity among the IT staff has increased; and the university incurred no additional technology costs for hardware, software, or changes to the network topology.

Improved Security

By implementing a Server and Domain Isolation solution using IPsec, UVV now has far better security throughout its entire network. Access to network resources is now highly regulated because access is only granted to trusted computer systems (that is, members of the UVV Active Directory structure) with the appropriate IPsec policies and credentials.

Server and Domain Isolation also enabled UVV to isolate the server systems on which the management application source code resides, providing an additional layer of security that didn't exist before within the university's faculty and staff network.

Lower Operating Cost That Facilitates Growth

For UVV, cost was a primary concern in finding an acceptable solution. The university was growing so quickly that it could not afford to continue expanding and maintaining two separate networks. In addition, the IT group didn't have the resources to build solutions that departments needed. "One UVV department asked us to build a new lab for it," says

Immaginario. “This would have required us to install fiber-optic cables, purchase a new network switch, set new server connections, and so on. To do so would have been prohibitively expensive.”

But implementing the Server and Domain Isolation solution based on IPsec essentially cost the university nothing. Instead, it allowed UVV to maximize its investments in Microsoft software because it was already running Windows Server 2003 on its server systems. And because IPsec is implemented at the network layer, the IT group didn’t have to purchase new hardware or software, or make any changes to the university’s applications or network topology. “With IPsec, now we can implement this project with wider security coverage—as a new wireless network—for professors and the students at all UVV campuses,” says Immaginario.

Improved Productivity

The Server and Domain Isolation solution enabled the IT group to consolidate the two networks into a single network infrastructure, which makes it much easier—and far less expensive—for the IT group to manage. Before, the IT staff had to manage each network individually. “With Server and Domain Isolation, we can manage the entire environment from a single console now,” says Immaginario. “Now we can just use a drag-and-drop operation to group departments together, then isolate that group, and it’s done.” Because managing the network takes far less time now, Immaginario expects productivity to increase dramatically. He already has plans to restructure the IT staff in 2006 so that team members can devote more time to developing solutions for the university.

“Server and Domain Isolation is an amazing solution. We already had all the tools, like IPsec, in our environment; we just didn’t use them before. Once we had time to study and to plan the IPsec solution, we did it quickly, more securely, and at no additional cost to the university,” says Immaginario. “When we tell other universities that they can improve security at no additional cost, no one believes it! Many of them have the same tools as well, they just don’t know about this IPsec-based solution yet.”

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